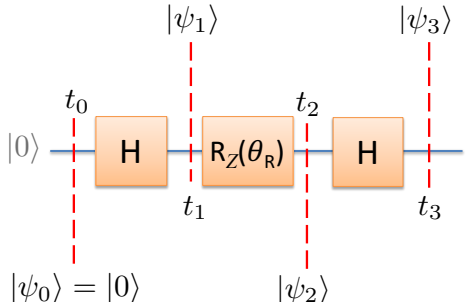
MULT20015 – Practice Questions

Please note: this list of question is not exhaustive in terms of syllabus coverage, and questions are not necessarily of equal difficulty and/or length.

1. Consider the sequence of gates shown below:



where the rotation $R_Z(\theta_\mathrm{R})$ is defined with zero global phase and is given by:

$$R_Z(\theta_\mathrm{R}) = \begin{bmatrix} \cos\frac{\theta_\mathrm{R}}{2} - i\sin\frac{\theta_\mathrm{R}}{2} & 0 \\ 0 & \cos\frac{\theta_\mathrm{R}}{2} + i\sin\frac{\theta_\mathrm{R}}{2} \end{bmatrix}$$

(a) Determine the states $|\psi_{1\to 3}\rangle$ at each time step in matrix representation keeping the rotation angle unspecified.

(b) Writing the final state in ket notation as $|\psi_3\rangle = a_0|0\rangle + a_1|1\rangle$:

(i) Write down the amplitudes a_0 and a_1.

(ii) Convert the state to (QUI) polar notation form: $|\psi_3\rangle = |a_0|e^{i\theta_0}|0\rangle + |a_1|e^{i\theta_1}|1\rangle$.

(iii) Convert to "Bloch sphere" form $|\psi_3\rangle = e^{i\theta_0}\left(\cos\frac{\theta_\mathrm{B}}{2}|0\rangle + \sin\frac{\theta_\mathrm{B}}{2}e^{i\phi_B}|1\rangle\right)$, and show that for this state the $\theta_0 = 0$ and Bloch angles θ_B and ϕ_B are θ_R and $-\pi/2$ repsectively.

2. (from 2020 exam)

Single-qubit rotations about the X, Y and Z axes are given by the following expressions in matrix notation:

$$R_X(\theta_\mathrm{R}) = \begin{bmatrix} \cos\frac{\theta_\mathrm{R}}{2} & -i\sin\frac{\theta_\mathrm{R}}{2} \\ -i\sin\frac{\theta_\mathrm{R}}{2} & \cos\frac{\theta_\mathrm{R}}{2} \end{bmatrix},$$

$$R_Y(\theta_\mathrm{R}) = \begin{bmatrix} \cos\frac{\theta_\mathrm{R}}{2} & -\sin\frac{\theta_\mathrm{R}}{2} \\ \sin\frac{\theta_\mathrm{R}}{2} & \cos\frac{\theta_\mathrm{R}}{2} \end{bmatrix},$$

$$R_Z(\theta_\mathrm{R}) = \begin{bmatrix} \cos\frac{\theta_\mathrm{R}}{2} - i\sin\frac{\theta_\mathrm{R}}{2} & 0 \\ 0 & \cos\frac{\theta_\mathrm{R}}{2} + i\sin\frac{\theta_\mathrm{R}}{2} \end{bmatrix}.$$

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(a) Starting from an arbitrary state $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$, where a_0 and a_1 are complex amplitudes, determine the amplitudes a_0' and a_1' in the final state $|\psi'\rangle = a_0'|0\rangle + a_1'|1\rangle$, obtained when the following sequence of operations is applied: $|\psi'\rangle = R_Y(\pi/2)R_Z(\pi/2)R_X(\pi/2)|\psi\rangle$. Determine the probabilities of measuring "0" and "1" in the state $|\psi'\rangle$. Show your working.

(b) For a starting state with $a_0 = a_1 = 1/\sqrt{2}$, determine the final state as expressed in polar form $|\psi'\rangle = |a_0'|e^{i\theta_0'}|0\rangle + |a_1'|e^{i\theta_1'}|1\rangle$ with the amplitude phase angles expressed as multiples of π. Convert the states $|\psi\rangle$ and $|\psi'\rangle$ to "Bloch sphere" form, and plot on the Bloch sphere. Show your working.

(c) For the example in (b) above, use geometric arguments on the Bloch sphere (including diagrams as necessary) to explain how the individual rotation operations take the initial state $|\psi\rangle$ to the final state $|\psi'\rangle$. Show your working.

3. Consider the single qubit state $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$, where a_0 and a_1 are complex amplitudes to be determined.

(a) Describe what happens when the qubit is measured (in the "computational" Z-basis). If the qubit was repeatedly initialised in this state and measured, how should the results be interpreted?

(b) On a quantum computer comprising n qubits, each qubit is independently initialised in the state $|\psi\rangle$ and subsequently measured. The process is repeated many times, producing a probability distribution over the binary numbers represented by the register. In terms of the amplitudes, a_0 and a_1, estimate the probability of obtaining the binary string corresponding to the integer $2^{n-1}+1$.

(c) Transform the single qubit state $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$, into the X-basis $\{|+\rangle, |-\rangle\}$ (where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$), and describe the results if the qubit, repeatedly initialised in $|\psi\rangle$, was measured in the X-basis.

(d) In terms of a quantum circuit, explain how measurements would be performed in the X-basis.

4. Alice and Bob would like to establish a secret key quantum key distribution.

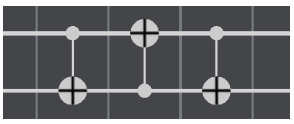
(a) If Alice sends a qubit in the x-basis, and would like to send the bit 0, which quantum state does she send?

(b) If Eve, the eavesdropper chooses to measure this qubit in the z-basis, what are her possible measurement probabilities and results?

(c) Eve sends on the qubit in the z-basis. Bob chooses to measure in the x-basis. What are his possible measurement results, and probabilities?

(d) Bob happens to measure "1". How might this let Alice and Bob know that there is an eavesdropper on the line?

5. Show that the combination of three CNOT gates shown in the circuit below,



acts as a SWAP gate for any two-qubit input state.

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6. Find an expression in matrix form for the single qubit gate operation G that produces the state $|\psi\rangle = \frac{1}{\sqrt{2}}e^{-i\pi/2}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ starting from the initial state $|0\rangle$. You can assume the unitary matrix for G and can be described by the parameterisation $G = \begin{bmatrix} a & b \\ -b^* & a^* \end{bmatrix}$, with $\det(G) = 1$.

Hence, construct a circuit that produces the two qubit entangled state $\frac{1}{\sqrt{2}}e^{-i\pi/2}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$ starting from the initial state $|00\rangle$.

7. (from 2020 exam)

Consider the following scenario. Two qubits Q1 and Q2 both initialised in the $|0\rangle$ state are prepared in an entangled state using a two-qubit entanglement-preparation circuit:

$$|\psi\rangle = \frac{1}{\sqrt{2}}[|00\rangle - |11\rangle],$$

where in the kets Q1 is the first listed and Q2 is the second. Q1 is sent to Zoe, while Q2 is sent to Alex. Alex can perform single-qubit operations on qubit Q2, corresponding to one of the following scenarios, before sending it on to Zoe:

Scenario 1: Alex performs no operations.
Scenario 2: Alex performs a X-gate.
Scenario 3: Alex performs a Z-gate.
Scenario 4: Alex performs a X-gate followed by a Z-gate.

Zoe then performs in the inverse of the entanglement-preparation circuit and measures both qubits.

(a) Write down the effective quantum circuit corresponding to one of these scenarios. For each of the four scenarios determine the probabilities for the outcomes of Zoe's measurements.

(b) Discuss the information transferred between Alex and Zoe (in terms of both qubits and classical bits). Explain what is significant about this protocol, and the role entanglement plays.

8. Consider the function, f, for which

f(0) = 1
f(1) = 0
f(2) = 0
f(3) = 1

(a) Construct a quantum circuit to implement this function.

(b) Is this function balanced or constant?

(c) Show the circuit you could construct to implement the Deutsch-Josza algorithm and evaluate whether this function is constant or balanced.

(d) How would you know, after implementing the DJ algorithm, whether this function is balanced or constant?

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(e) How many applications of the function does it require?

9. The first few square numbers are 1, 4 and 9. Grover's algorithm is performed using a four qubit register.

(a) Design an oracle which marks the square numbers (ie. applies a -1 phase to each of these states). You may use multiply controlled gates.

(b) How many iterations of Grover's algorithm would be required? Show your working.

(c) How many iterations of Grover's algorithm would be required to find all the numbers less than one million using Grover's algorithm? Use the minimum number of qubits necessary.

10. Given two data sequences, D1 of size "n" and D2 of size "m". Suppose D1 and D2 are both sorted in ascending order. If D2 is a subsequence of D1, then all elements in D2 can be produced by removing elements from D1. This implies that there are indices $i_1 < \ldots < i_m$ such that D1[i] = D2[j] for each j.

(a) Design and describe a classical algorithm that detects if D2 is a subsequence of D1 and outputs the indices $i_1, \ldots, i_m$ if D2 is indeed a subsequence of D1.

(b) What is the elementary operation in your algorithm? What is the cost function?

(c) Find worst case Big-O cost.

11. (from 2020 exam)

Consider a function $f(x)$ where x is a n-bit input. The function returns "1" if there are strictly more bits set to zero in the input than there are bits set to one, otherwise it returns "0".

(a) Design a classical algorithm that implements $f(x)$ for an input size of n=3 bits, which would scale as $O(n)$ when the number of input bits, n, increases.

(b) Design a quantum circuit that implements $f(x)$ for x of size n=3 bits. Draw the quantum circuit using one-qubit gates, two-qubit gates and *just two* Toffoli gates. Explain its working. You may use ancilla qubits.

(c) Imagine that all you knew about the function, $f(x)$ was that it is either balanced or constant. Outline how a quantum algorithm could determine if the function you implemented in part (b) is balanced or constant? Explain your answer.

12. (from 2020 exam)

We wish to factor N=91, on a quantum computer using Shor's algorithm with a base of a=3. Let us imagine that we have two registers. The upper register has 5 qubits.

(a) After the inverse Fourier transform, the value y=27 is measured in the upper register. Based on this measurement, use the continued fractions approximation to find the period, r, of 3^x mod 91. Show your working.

(b) Given your answer for (a), find the non-trivial factors of 91. Show your working.

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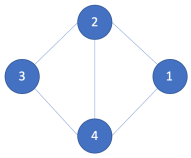
13. In the MAX-CUT problem, you are given a graph G consisting of vertices, V and edges E. The MAX-CUT problem you have to find a partition of the vertices V, such that there are the maximum number of edges connecting the two partitions.

(a) Given this graph:



Find the two partition which maximizes the number edges between the partitions.

(b) Write down a corresponding problem Hamiltonian, which translates this problem into an energy minimization problem.

(c) Which states correspond to the solution?

(d) Draw the circuit to construct the trial state, for one round of QAOA.

14. Consider a two asset portfolio optimization problem where the expected returns of assets A and B are 6% and 8%, respectively, the variance of assets A and B are 0.004 and 0.007 respectively (their covariance is unknown). The risk penalty, budget and budget penalty are 5, $2, and 10 respectively. The translation of this problem to QAOA is via introducing binary variables to denote a dollar invested in an asset.

The expression to optimise in the portfolio optimisation problem is:

$$\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$$

subject to $\sum_{i=1}^{n} w_i = 1$

(a) You are told that the equal weighted portfolio (i.e. a portfolio with weights of asset A and B as 0.5) has a risk of 0.00175. Complete the table below by finding the covariance between asset A and B.

	Asset A	Asset B
Asset A	0.004	x
Asset B	x	0.007

(b) Complete the table below by calculating the return, risk, and the value of the portfolio for the given weights in the first column.

Weights	Portfolio return	Portfolio risk	Portfolio value
$W_A = 0.5$, $W_B = 0.5$		0.00175	
$W_A = 1$, $W_B = 0$			

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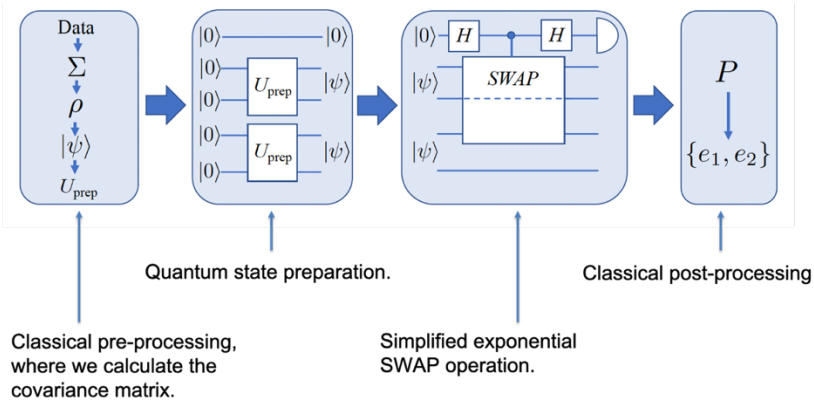
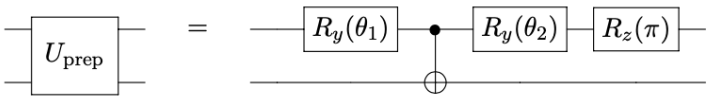
(c) If you are told that the weights maximise the portfolio value are $W_A = 0.5$, $W_B = 0.5$, which quantum states will you expect to have the highest probabilities?

15. In lectures we studied quantum principal component analysis algorithm which involves the following steps:



In the practice class we saw that for a simple 2×2 covariance matrix we could use the following circuit for the U_{prep} step:



(a) Suppose we are given the following 2-dimensional dataset:

Number of years of education = {10, 14, 16, 20, 24}
Salary per year = {65000, 71000, 78000, 81000, 102000}

Find covariance matrix for the above dataset. Calculate the eigenvalue corresponding to the principal component by diagonalising the covariance matrix.

(b) Based on the diagram above, find ρ and U_{prep}.

(c) Implement the QPCA circuit in QUI and find eigenvalues from the outcome of QPCA circuit.

(d) How do answers from part (a) and part (c) compare?

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Solutions:

Question 1

1 (a)

$$|\psi_0\rangle = |0\rangle = \begin{pmatrix}1\\0\end{pmatrix},\ |\psi_1\rangle = (1/\sqrt{2})\begin{pmatrix}1\\1\end{pmatrix}, |\psi_2\rangle = (1/\sqrt{2})\begin{pmatrix}\cos\frac{\theta_R}{2} - i\sin\frac{\theta_R}{2}\\ \cos\frac{\theta_R}{2} + i\sin\frac{\theta_R}{2}\end{pmatrix}, |\psi_3\rangle = \begin{pmatrix}\cos\frac{\theta_R}{2}\\ -i\sin\frac{\theta_R}{2}\end{pmatrix}$$

1 (b) (i) $|\psi_3\rangle = \begin{pmatrix}\cos\frac{\theta_R}{2}\\ -i\sin\frac{\theta_R}{2}\end{pmatrix} = \cos\frac{\theta_R}{2}\ |0\rangle - i\sin\frac{\theta_R}{2}\ |1\rangle$

1 (b) (ii) $|\psi_3\rangle = |a_0|e^{i\theta_0}|0\rangle + |a_1|e^{i\theta_1}|1\rangle$

$|a_0| = \cos\frac{\theta_R}{2}, \theta_0 = 0;\ |a_1| = \sin\frac{\theta_R}{2}, \theta_1 = -\pi/2$

1 (b) (iii) $|\psi_3\rangle = e^{i\theta_0}\left(\cos\frac{\theta_B}{2}|0\rangle + \sin\frac{\theta_B}{2}e^{i\phi_B}|1\rangle\right) \leftrightarrow |\psi_3\rangle = \cos\frac{\theta_R}{2}|0\rangle + \sin\frac{\theta_R}{2}e^{-i\pi/2}|1\rangle$

Question 2

(a)

$$\begin{aligned}
|\psi'\rangle &= R_Y(\pi/2)R_Z(\pi/2)R_X(\pi/2)\,|\psi\rangle \\
&= \frac{1}{2\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}1-i & 0\\ 0 & 1+i\end{bmatrix}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix}\begin{bmatrix}a_0\\ a_1\end{bmatrix} \\
&= \frac{-(1+i)}{\sqrt{2}}a_1\,|0\rangle + \frac{1-i}{\sqrt{2}}a_0\,|1\rangle\,. \\
\Pr(0) &= |a_1|^2, \\
\Pr(1) &= |a_0|^2.
\end{aligned}$$

(b) Here $|\psi'\rangle$ can be worked out by substituting into part (a), or through rotations on the Bloch sphere (but check you have the correct global phase).

$$\begin{aligned}
|\psi'\rangle &= \frac{1}{\sqrt{2}}e^{-i3\pi/4}\,|0\rangle + \frac{1}{\sqrt{2}}e^{-i\pi/4}\,|1\rangle \\
&= e^{-i3\pi/4}\left(\cos\left(\frac{\pi/2}{2}\right)|0\rangle + e^{i\pi/2}\sin\left(\frac{\pi/2}{2}\right)|1\rangle\right).
\end{aligned}$$

A diagram of the Bloch sphere with $|\psi'\rangle$ at $(0,1,0)$.

(c) Using Bloch sphere diagram(s): starting from $(1,0,0)$, rotate by $\pi/2$ around the X axis (no-op), then $\pi/2$ around the Z axis to give the point at $(0,1,0)$, then $\pi/2$ around the Y axis (no-op).

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Question 3

3 (a) $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$

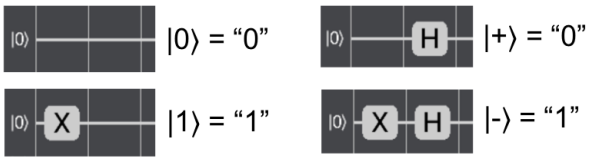
Z-basis measurement will produce either "0" (collapse $|\psi\rangle$ into $|0\rangle$), OR "1" (collapse $|\psi\rangle$ into $|1\rangle$), according to the probabilities given by $|a_0|^2$ or $|a_1|^2$ respectively.

Repeated initialisation and measurement -> build a histogram of results "0" and "1" corresponding to the probabilities $|a_0|^2$ or $|a_1|^2$.

3 (b) Probability of measuring strings of bits = |amplitude(state <-> a given bit string)|^2. The bit string for the integer $2^{n-1}+1$ is 100…0001, with amplitude $(a_1)^2(a_0)^{n-2}$. The corresponding probability to obtain this state is $|(a_1)^2(a_0)^{n-2}|^2$.

3 (c) Transformation gives: $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$ -> $|\psi\rangle = \frac{a_0+a_1}{\sqrt{2}}\ |+\rangle + \frac{a_0-a_1}{\sqrt{2}}\ |-\rangle$
-> Probability of measuring "+" is $|\frac{a_0+a_1}{\sqrt{2}}|^2$, and probability of measuring "-" is $|\frac{a_0-a_1}{\sqrt{2}}|^2$.

3 (d) Transformation between measurements in Z and X bases is given by:



Question 4

4 (a) Alice should send the state

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

4 (b) Eve, measuring in the computational basis will measure either measure "0" with probability

$$P_0 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

Or measure "1" with probability

$$P_1 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

4 (c) Bob is measuring in the x-basis, and so will apply a Hadamard gate to whatever Eve sends. Regardless of whether she sends in the z-basis, Bob will measure the "+" (0) or "-" (1) states, each with probability

$$P_+ = P_- = \frac{1}{2}$$

4 (d) After they have exchanged qubits, Alice and Bob publicly announce the basis they sent and measured in. They discard any that have different bases. Then for a some of the remaining qubits they compare what was sent (by Alice) and what was received (by Bob). Without an eavesdropper, these results should be the same.

Since in this case both Alice and Bob sent and received the qubit in the same basis (the x-basis) they do not discard the qubit, and the data which Alice sent should be what Bob received. However, in this case Alice sent

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the data "0", and Bob received the data "1". If Alice and Bob compare the data which was sent to what was received (for this qubit) they will therefore realise there is an eavesdropper on the line.

Question 5

5. Write the general two qubit state as (qubit 1 = top, qubit 2 = bottom):

$a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$

After 1st CNOT controlled on quibt-1 we obtain:

$$a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|11\rangle + a_{11}|10\rangle$$

After 2nd CNOT controlled on quibt-2 we obtain:

$$a_{00}|00\rangle + a_{01}|11\rangle + a_{10}|01\rangle + a_{11}|10\rangle$$

After 3rd CNOT controlled on quibt-1 we obtain:

$$a_{00}|00\rangle + a_{01}|10\rangle + a_{10}|01\rangle + a_{11}|11\rangle$$

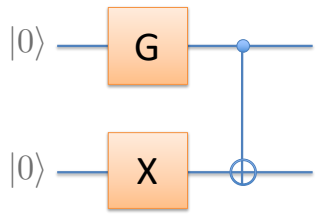
i.e. at output, the assignments in the kets have been swapped with respect to the input state.

Question 6

Apply G to the $|0\rangle$ state to produce the required state, and solve for the parameters a and b in G to obtain:

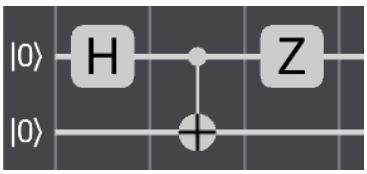
$$G = \frac{1}{\sqrt{2}}\begin{bmatrix} e^{-i\pi/2} & -1 \\ 1 & e^{i\pi/2} \end{bmatrix}$$

The circuit to produce the required entangled state proceeds as follows:



Question 7

7 (a)
One circuit, among many, which implements the preparation of $|\psi\rangle$ is:



The inverse of the entanglement generation circuit is:

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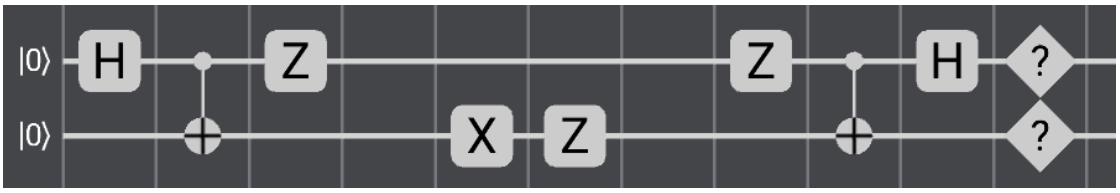
If scenario 4 was chosen then full circuit is:



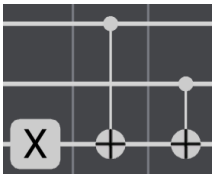
The measurement probabilities for each of the scenarios are:

Scenario	Pr(00)	Pr(01)	Pr(10)	Pr(11)
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

7 (b)
During the procedure Alex sends 1 qubit to Zoe. After performing the inverse of the entanglement-preparation circuit and measurement Zoe can deduce exactly which of the 4 scenarios Alex chose to implement. The 4 possible scenarios correspond to the 4 possible states 2 classical bits can occupy at any given instant (00, 01, 10, 11). This process has sent 2 classical bits of information from Alex to Zoe. This procedure relies on entanglement between Alex's qubit with Zoe's qubit existing while Alex is in possession of his qubit.

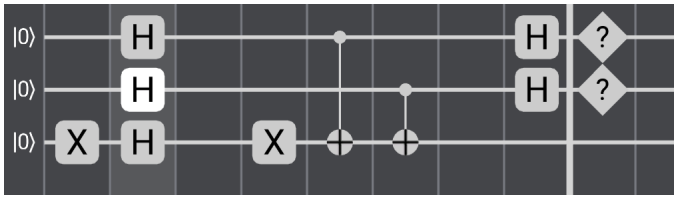
Question 8

8 (a) The function, f, can be constructed with the following circuit:



8 (b) The function, f, is a <u>balanced</u> function.

8 (c) The Deutsch-Josza algorithm, for this function, can be constructed using the following circuit:



8 (d) After running the algorithm a non-zero measurement result is obtained when measuring the upper register:

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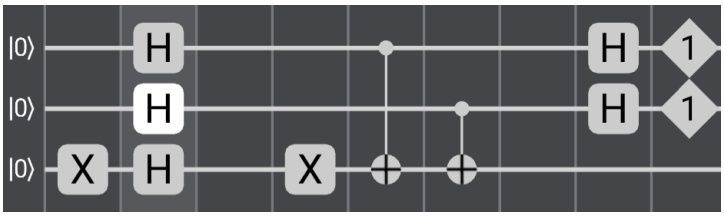
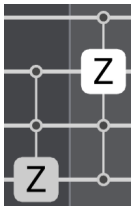


This indicates that the function is <u>balanced</u>.

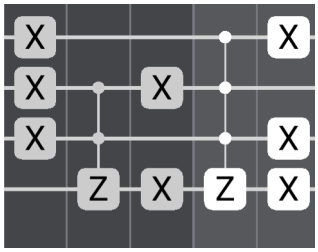
8 (e) Only <u>one</u> application of the function is required.

Question 9

9 (a) The following circuit gives one (of many) possibility for the oracle:



Or (without control-on-0 gates):



9 (b) In the geometric picture:

$$sin\,\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

where the total number of solutions is

$$M = 3$$

And the dimension is

$$N = 2^4 = 16$$

So, for this problem

$$sin\,\theta = \frac{\sqrt{3}}{\sqrt{16}}$$
$$\therefore \theta = 0.448$$

Ideally, after k iterations we would like an angle of $\pi/2$. In that case,

$$(2k+1)\theta = \frac{\pi}{2}$$

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$$\therefore k = \frac{1}{2}\left(\frac{\pi}{2\theta} - 1\right)$$

Substituting in theta gives:

$$k = 1.25$$

This search would require therefore require <u>just one</u> iteration.

9 (c) If there are M squares less than 1,000,000 then

$$M = \left\lfloor \sqrt{999{,}999} \right\rfloor$$
$$= 999$$

If there are n qubits required,

$$n = \lceil \log_2 1{,}000{,}000 \rceil$$
$$= 20$$

giving

$$N = 2^n = 1048576$$

Again,

$$\sin\theta = \frac{\sqrt{999}}{\sqrt{1048576}}$$
$$\therefore \theta = 0.0308$$

Ideally, after k iterations we would like an angle of $\pi/2$. In that case,

$$(2k+1)\theta = \frac{\pi}{2}$$
$$\therefore k = \frac{1}{2}\left(\frac{\pi}{2\theta} - 1\right)$$

Substituting in theta gives:

$$k = 24.9$$

Therefore, we would require approximately <u>25 iterations</u>.

Question 10

10 (a) Suppose D1 is a sequence of n numbers with indices $i_1, i_2, \ldots i_n$, and D2 is a sequence of m numbers with indices $j_1, j_2, \ldots, j_m$. The algorithm will use binary search to locate D2[j_1] in D1. If no match found, the algorithm concludes that D2 is not a subsequence of D1 and exits. If a match found say at index i_k, then the algorithm will apply binary search again to find second element of D2[j_2] in D1 subarray defined by indices i_{k+1} to i_n. The algorithm iteratively repeats itself until all elements of D2 are located in D1, in which case the answer is yes, D2 is a subsequence of D1 and the found indices are returned. If no match is found for any single element of D2, then the answer is: D2 is not a subsequence of D1.

10 (b) The elementary operation is the comparison of an element from D2 with an element in D1. Recall that the cost of binary search is $\log_2(n)$. Locating the first element of D2 in D1 would require $\log_2(n)$ comparisons under worst case. Suppose this element is at the centre of D1 [worst case], then the second element of D2 will have to be searched in (n/2) elements of D1, requiring $\log_2(n/2)$ comparisons, and so on. The total cost function is: $\log_2 n + \log_2 (n/2) + \log_2 (n/4) + \ldots \log_2 (n/2^{m-1})$.

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10 (c) The asymptotic cost of the function is $O(\log_2(n^m))$.

Question 11

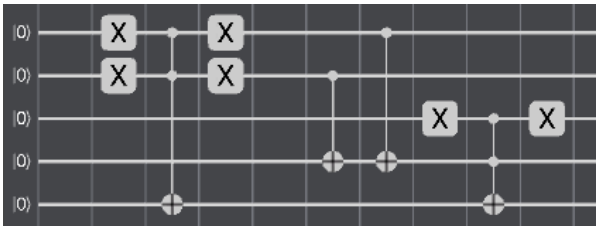
11 (a)
Classical algorithm (among others) for n bit input:
Start a counter at 0. Begin an iteration procedure, with a maximum number of iterations n, and that stops if the counter ever reaches a number greater than n/2. Look at a unique bit of the input each iteration and increment the counter by 1 if the bit was 0. Return "1" if the counter was greater than n/2 and "0" otherwise. If the value of the counter is greater than n/2 then there must have been more than half of the bits set to 0.

11 (b)
One solution (among many):

A good approach is to look at other ways of describing the function.

x	000	001	010	011	100	101	110	111
$f(x)$	1	1	1	0	1	0	0	0

One thing to notice is that if the first two bits are zero then the output is 1. This can be implemented with a Toffoli gate decorated with X gates. This hands 2 of the 4 states that need to return 1. It can be observed that the other two states 010 and 100 have the quality that they are the only states that have different first and second bit values and a third bit value of 0. We can use an ancilla and CNOT gates to store whether the first two bits are the same. We can use a Toffoli to change the output when the first two bits are different, and the last bit is zero. The full circuit, where the first 3 qubits are the input, the 4th qubit is an ancilla and the 5th qubit is the output, is:



11 (c)
The 3-bit function is balanced. The Deutsch-Josza algorithm can implemented, and a nonzero measurement result will confirm the function is balanced.

Question 12.

12 (a)
The continued fraction expansion of $\frac{27}{2^6} = \frac{27}{32}$ is:

$$\cfrac{1}{1+\cfrac{1}{5+\cfrac{1}{2+\cfrac{1}{2}}}}$$

This can be truncated as $\frac{1}{1}$ to give a denominator 1. We find $3^1 = 3 \neq 1 \bmod 91$ so we need to truncate later.

Truncating as $\frac{1}{1+\frac{1}{5}}$ evaluates to $\frac{5}{6}$. Verify that $3^6 = 729 = 1 \bmod 91$. Thus, the period is $r = 6$.

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12 (b)
The factors are found with $\gcd\left(a^{r/2} \pm 1, n\right) = \gcd\left(3^{6/2} \pm 1, 91\right) = \gcd(3^3 \pm 1, 91)$. You should find factors 7 and 13.

Question 13

13 (a) The solution partitions vertices into

$$\{1,3\} \text{ and } \{2,4\}.$$

13 (b) If each vertex corresponds to a qubit, and the state of the qubit represents which partition that vertex is in then we can construct a Hamiltonian which adds an energy of -1 for each edge connecting between partitions:

$$H = \sum_{\langle i,j \rangle \in E} \frac{Z_i Z_j - 1}{2}$$

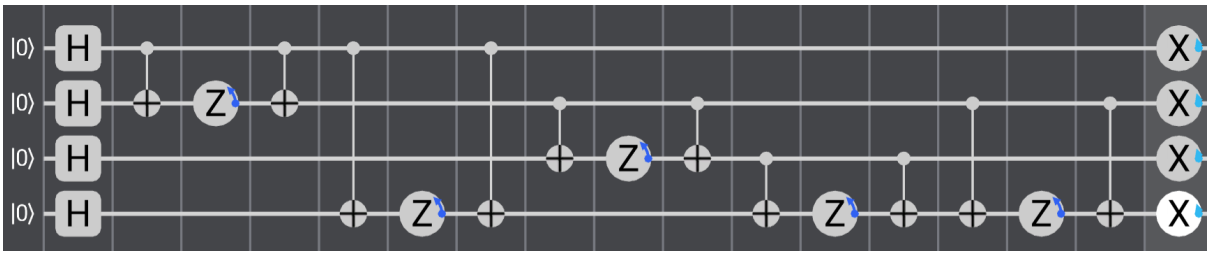
where the sum is over all edges in the graph. In our case that gives,

$$H = \frac{1}{2}(Z_1Z_2 + Z_1Z_4 + Z_2Z_3 + Z_3Z_4 + Z_2Z_4 - 5)$$

13 (c) Solution states are

$$|0101\rangle, |1010\rangle$$

13 (d)



Where each of the z-rotation angles are given by $-\alpha/2$, and each of the x-rotation angles are by an angle of β.

Question 14

14 (a) We have one equation (portfolio risk) and one unknown (covariance).

$(0.5^2)0.004 + (0.5^2)0.007 + 2(0.5)(0.5)x = 0.00175$
Solving for x we get x = -0.002

14 (b) Values in the table:

Weights	Portfolio return	Portfolio risk	Portfolio value
$w_A = 0.5$, $w_B = 0.5$	7%	0.175 or 0.00175	6.125 or 0.06125
$w_A = 1$, $w_B = 0$	6%	0.4 or 0.004	4.0 or 0.04

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14 (c) In this case we expect to see 4 (equivalent) states: $|1001\rangle$, $|0101\rangle$, $|1001\rangle$, $|0101\rangle$.

Question 15

15 (a) The given dataset is:

Number of years of education = {10, 14, 16, 20, 24}
Salary per year = {65000, 71000, 78000, 81000, 102000}

After standardisation:

X = Number of years of education = {10, 14, 16, 20, 24}
Y = Salary per year = {6.5, 7.1, 7.8, 8.1, 10.2}

E[X] = 16.8
E[Y] = 7.94

Covariance = $\begin{bmatrix} 29.2 & 7.26 \\ 7.26 & 1.983 \end{bmatrix}$

Eigenvalues = 0.1675, 31.0155

The Eigenvalue corresponding to the principal component is 31.0155

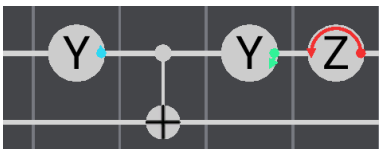
15 (b)

$\rho = \begin{bmatrix} 0.9364 & 0.2328 \\ 0.2328 & 0.0636 \end{bmatrix}$

$$\begin{bmatrix} 0.9364 & 0.2328 \\ 0.2328 & 0.0636 \end{bmatrix} = \begin{pmatrix} \frac{1}{2}(1 + Cos(\theta_1)\, Cos(\theta_2)) & -\frac{1}{2} Cos(\theta_1)\, Sin(\theta_2) \\ -\frac{1}{2} Cos(\theta_1)\, Sin(\theta_2) & \frac{1}{2}(1 - Cos(\theta_1)\, Cos(\theta_2)) \end{pmatrix}.$$

One solution to these equations is: $\theta_1 = 0.1471,\ \theta_2 = -0.4901$

The circuit for U_{prep} is:



15 (c) The QUI circuit is:

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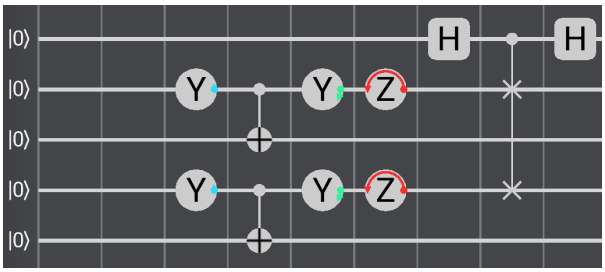

P = 0.986

Trace(Covariance) = 31.183

$$\lambda_1 = Trace[Covariance]\frac{\left(1+\sqrt{1-2(1-P)}\right)}{2}$$
$$\lambda_2 = Trace[Covariance]\frac{\left(1-\sqrt{1-2(1-P)}\right)}{2},$$

$\lambda_1 = 30.9616$

$\lambda_2 = 0.218$

15 (d)

The estimated eigenvalues from quantum circuit are close but slightly different from part (a).

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